



McKetta Technical Electives

2020–2022, 2022–2024, 2024–2026, and 2026–2028 catalogs

Requirements. Choose one or two technical option areas. Complete four electives in one area, or two electives in each of two areas. At least two electives must be engineering courses, and at least one of those must be a CHE course.

Technical option areas

Area 1 Process System and Product Engineering

Area 2 Materials Engineering

Area 3 Environmental Engineering

Area 4A Cellular and Bioprocess Engineering

Area 4B Biomedical Engineering

Area 5 Energy Technologies

Area 6 Engineering Economics and Business Leadership

Courses are grouped as ChE, other engineering, and non-engineering. A checkmark identifies an approved technical option area for the listed course or topic.

ChE courses	Area 1	Area 2	Area 3	Area 4A	Area 4B	Area 5	Area 6
CHE 322M , <i>Molecular Thermodynamics</i>		✓					
CHE 323 , <i>Chemical Engineering for Microelectronics</i>		✓				✓	
CHE 337 , <i>Quantitative Analysis of Cellular and Molecular Biology (CHE 379)</i>	✓			✓	✓		
CHE 339 , <i>Introduction to Biochemical Engineering</i>				✓	✓	✓	
CHE 339P , <i>Introduction to Biological Physics</i>				✓	✓		
CHE 339T , <i>Cell and Tissue Engineering</i>					✓		
CHE 341 , <i>Design for Environment</i>	✓		✓			✓	
CHE 342 , <i>Chemical Engineering Economics and Business Analysis</i>	✓						✓
CHE 343 , <i>Molecular Simulation of Materials</i>	✓	✓					
CHE 344 , <i>Introduction to Therapeutics (CHE 379)</i>				✓	✓		
CHE 355 , <i>Introduction to Polymers (CH 367L)</i>		✓			✓	✓	
CHE 356 , <i>Optimization: Theory and Practice</i>	✓						✓
CHE 357 , <i>Technology and Its Impact on the Environment</i>			✓			✓	
CHE 359 , <i>Energy Technology and Policy (M E 379M)</i>			✓			✓	
CHE 364S , <i>Chemical Process Safety</i>	✓	✓	✓	✓	✓	✓	✓
CHE 376K , <i>Process Evaluation and Quality Control</i>	✓		✓				
CHE 377K/L , <i>Undergraduate Research Project</i>	✓	✓	✓	✓	✓	✓	✓
CHE 379 , <i>Adv Analysis for Chem Engineer</i>	✓						
CHE 379 , <i>Advance Nanotechnology & Innovation for Beginners (study abroad)</i>	✓	✓	✓	✓	✓	✓	✓
CHE 379 , <i>Advance Polymer Synthesis</i>		✓					
CHE 379 , <i>Atmospheric Chemistry and Physics</i>			✓			✓	

ChE courses	Area 1	Area 2	Area 3	Area 4A	Area 4B	Area 5	Area 6
CHE 379, Atmospheric Physiochemical Processes			✓				
CHE 379, Biochemical, Cellular, and Metabolic Engineering	✓			✓	✓	✓	
CHE 379, C*star Elective		✓	✓			✓	
CHE 379, Catalyst for Sustainable Future		✓	✓			✓	
CHE 379, Chem for a Sustainable Future			✓			✓	
CHE 379, Computation Methods with Application to Materials		✓					
CHE 379, Data Analytics Apps in ChE	✓						
CHE 379, Dynamics in Ecology and the Environment	✓		✓	✓	✓		
CHE 379, Electrochemistry for Chemical Engineering	✓	✓				✓	
CHE 379, Energetic Separations	✓		✓			✓	
CHE 379, Energy Transition for a Sustainable Future						✓	
CHE 379, Energy and the Environment			✓			✓	
CHE 379, Engineering Global Health (study abroad)	✓	✓	✓	✓	✓	✓	✓
CHE 379, Engineering and Public Policy			✓			✓	✓
CHE 379, Entrepreneurship	✓	✓	✓	✓	✓	✓	✓
CHE 379, From Data to Decisions	✓	✓				✓	✓
CHE 379, Fundamentals & Applications of Cellular Regulation				✓	✓		
CHE 379, Global Carbon Monitoring Systems (Cross-Listed PGE 379)			✓			✓	
CHE 379, Greenhouse Gas Control Technologies	✓		✓			✓	
CHE 379, International Perspectives on BME Design					✓		
CHE 379, Machine Learning in Biology				✓	✓		
CHE 379, Materials Physics	✓	✓	✓			✓	
CHE 379, Modeling of Nanopatterning Processes	✓	✓					
CHE 379, Modeling the Earth's Systems	✓		✓			✓	
CHE 379, Modern Biointerface Sci & Engr				✓	✓		
CHE 379, Molecular Biophysics: MSRMNTS/MTHDS				✓	✓		
CHE 379, Nanomaterials Chemistry & Engineering	✓	✓	✓	✓	✓	✓	
CHE 379, Nanotechnology Innovation (study abroad)	✓	✓	✓	✓	✓	✓	✓
CHE 379, Photovoltaics	✓	✓				✓	
CHE 379, Physics		✓					
CHE 379, Polymer Kinetics and Reaction Engineering Polymer		✓					
CHE 379, Polymer Physics					✓		
CHE 379, Product Design and Commercialization	✓						✓
CHE 379, Renewable Energy			✓			✓	
CHE 379, Soft Matter Science		✓					
CHE 379, Trends in ChE Research	✓	✓	✓			✓	
CHE 679HA + CHE 679HB, Undergraduate honors thesis	✓	✓	✓	✓	✓	✓	✓

Other engineering courses	Area 1	Area 2	Area 3	Area 4A	Area 4B	Area 5	Area 6
ARE 323K , <i>Project Management and Economics</i>	✓						✓
BME 352 , <i>Engineering Biomaterials</i>					✓		
BME 353 , <i>Transport Phenomena in Living Systems</i>					✓		
BME 365R , <i>Quantitative Engineering Physiology I</i>					✓		
BME 377T , <i>Delivery of Therapeutic Agents</i>					✓		
BME 377T , <i>Global Health Innovation</i>				✓	✓		
BME 377T , <i>Molecular Sensors</i>					✓		
C E 341 , <i>Introduction to Environmental Engineering</i>			✓			✓	
C E 342 , <i>Water and Wastewater Treatments</i>			✓				
C E 364 , <i>Design of Wastewater and Water Treatment Facilities</i>			✓				
C E 369L , <i>Air Pollution Engineering</i>			✓				
C E 370K , <i>Environmental Sampling and Analysis</i>			✓				
ECE 334K , <i>Quantum Theory of Electronic Materials (E E 334K)</i>		✓					
ECE 339 , <i>Solid-State Electronic Devices (E E 339)</i>		✓				✓	
ECE 370K , <i>Computer Control Systems (E E 370K)</i>	✓						
ECE 374K , <i>Biomedical Electronic Instrument Design (E E 374K)</i>					✓		
ECE 379K , <i>Statistical Quality Control (E E 379K)</i>	✓						
E S 321G , <i>GLUE Undergrad Research</i>	✓	✓	✓	✓	✓	✓	✓
E S 369N , <i>Sustainability Issues in Engineering</i>						✓	
M E 334 , <i>Materials Engineering</i>		✓					
M E 335 , <i>Engineering Statistics</i>	✓						
M E 336P , <i>Nuclear Science and Engineering</i>		✓	✓			✓	
M E 337C , <i>Introduction to Nuclear Power System</i>						✓	
M E 337G , <i>Nuclear Safety and Security</i>						✓	
M E 348F , <i>Advance Mechatronics II</i>	✓						
M E 349 , <i>Corrosion Engineering</i>		✓					
M E 353 , <i>Engineering Finance</i>	✓						✓
M E 354 , <i>Introduction to Biomechanical Engineering</i>					✓		
M E 359 , <i>Materials Selection</i>		✓					
M E 361E , <i>Nuclear Operation and Reactor Engineering</i>						✓	
M E 363M , <i>Energy Technology and Policy</i>						✓	
M E 366L , <i>Operations Research Models</i>	✓						✓
M E 374F , <i>Fire Science</i>		✓					
M E 374S , <i>Solar Energy System Design</i>		✓	✓			✓	
M E 379M , <i>Concepts in Nuclear/Radiation Engineering</i>						✓	
PGE 301 , <i>Engineering, Energy, and the Environment</i>						✓	
PGE 323K , <i>Reservoir Engineering I: Primary Recovery</i>						✓	
PGE 365 , <i>Resources Economics and Valuation</i>						✓	

Other engineering courses	Area 1	Area 2	Area 3	Area 4A	Area 4B	Area 5	Area 6
PGE 379, Facilities Management						✓	
PGE 379, Geothermal Energy Technologies						✓	
PGE 430, Drilling and Well Completions						✓	

Non-engineering courses	Area 1	Area 2	Area 3	Area 4A	Area 4B	Area 5	Area 6
ACC 310F, Foundations of Accounting							✓
BCH 339F, Foundations of Biochemistry				✓	✓		
BCH 339J, Chemical and Synthetic Biology				✓	✓		
BCH 339M, Structure and Function of Molecular Machines				✓	✓		
BCH 339N, Systems Biology and Bioinformatics				✓	✓		
BCH 364C, Bioinformatics				✓	✓		
BCH 369, Fundamentals of Biochemistry				✓	✓		
BCH 370, Physical Methods of Biochemistry				✓	✓		
BIO 325, Genetics				✓	✓		
BIO 339, Metabolism and Biochemistry of Microorganism				✓			
CH 341, Special Topics in Laboratory Chemistry		✓					
CH 354, Quantum Chemistry and Spectroscopy		✓					
CH 354L, Physical Chemistry II		✓					
CH 368, The Structure and Function of DNA Materials		✓		✓		✓	
CH 376K, Advance Analytical Chemistry		✓					
ECO 304K, Introduction to Microeconomics							✓
ECO 304L, Introduction to Macroeconomics							✓
ECO 328, Industrial Organization							✓
ECO 339K, International Trade and Investment							✓
ECO 351K, Current Issues to Business Economics							✓
FIN 357, International Business Operation							✓
GEO 303E, Earth in 2100			✓				
I B 378, International Business Operations							✓
INB 365S, Human Systems Physiology (BIO 365S)					✓		
MAN 327, Innovation & Entrepreneurship							✓
MBS 320, Cell Biology (BIO 320)					✓		
MBS 326R, General Microbiology (BIO 326R)				✓	✓		
MIS 302F, Foundation Info Technology MGMT							✓
MKT 320F, Foundations of Marketing	✓						✓
MKT 460, Information and Analysis							✓
PHY 338K, Electronic Techniques		✓					
PHY 355, Modern Physics and Thermodynamics Introductory		✓					
PHY 375S, Solid-State Physics		✓					
Upper-division mathematics course (limit one)	✓						

Additional option requirements

Research. CHE 377K/L and E S 321G count in any area. A maximum of one research course may count as a technical elective.

Honors thesis. CHE 679HA and CHE 679HB together count as six hours of technical elective credit in any area. They may not be used as technical electives at the same time as CHE 377K/L or E S 321G.

Co-op. Three credit hours count as one technical elective in any area. Only three hours of co-op may count toward the technical area or areas.

Projects for Underserved Communities. Three terms count as one technical elective in any area.